

Ben Robison

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Education

University of Washington – BS in Electrical and Computer Engineering June 2028 (Expected)

- GPA: **3.8**; Dean's List (six quarters); Tau Beta Pi Engineering Honor Society

Skills

RTL/HDL: SystemVerilog, Verilog; RTL design, self-checking testbenches, FPGA synthesis, timing analysis

Programming: C, Python, Java, SQL, Git; embedded Linux

Hardware/EDA: ModelSim, Quartus Prime, DE1-SoC, Avalon-MM, KiCad, oscilloscope, soldering

Selected Hardware Projects

FPGA GEMM Accelerator github.com/Ae0lis/Systolic-Array

- Designed an 8×8 signed-INT8 GEMM accelerator in SystemVerilog using 64 parallel MAC units with INT32 accumulation; synthesized to 64 Cyclone V DSP blocks.
- Integrated the accelerator with the ARM HPS over Avalon-MM and wrote an embedded Linux C driver to load operands, control execution, and compare all 64 outputs against a software reference on a DE1-SoC.
- Computed one 8×8 -by- 8×8 matrix product in nine core cycles (180 ns at 50 MHz, excluding host transfers) with a 77.48 MHz post-fit worst-corner Fmax.
- Verified the RTL with self-checking testbenches and 1,000 randomized control-interface operations.

Line-Following Robot ae0lis.github.io/projects/line_robot

- Designed and assembled a 40×80 mm seven-channel photoresistor/LED sensor PCB in KiCad, using per-channel illumination and ambient-light shielding to improve line contrast.
- Mapped the sensor array to an Arduino control loop and tuned its PID gains, enabling the robot to complete the faculty-designed competition track.

Experience

Program Analyst, CBRE (Google Account) – San Jose, CA Aug 2026 – Present

Summer Associate, CBRE (Google Account) – San Jose, CA Jun 2026 – Aug 2026

- Rehired as a program analyst to improve asset mapping and identify data gaps supporting predictive maintenance efforts.
- Designed the SQL backend and dashboard frontend for a KPI display; presented the architecture to a senior Google stakeholder and secured approval for implementation.
- Audited an engineering report to guide facilities operations strategy, avoiding approximately \$10K in unnecessary work.

Embedded Systems Engineer, Husky Robotics @ UW – Seattle, WA Feb 2026 – Jun 2026

- Automated the rover's soil-sampling sequence in C on a Raspberry Pi, coordinating auger, lift, and collection-box actuation with time-of-flight fill detection and Hall-effect sensors for auger-column homing and dig-stop limits.
- Refactored the shared embedded-C codebase using conditional compilation and rewrote Makefile targets, eliminating duplicate entry-point conflicts and enabling subsystem reuse in automated routines.
- Built and tested PWM mappings and collection-box controls; supported rover electronics assembly through soldering, crimping, and wiring.

Technical Writer, Sensors, Energy, and Automation Lab (SEAL) @ UW – Seattle, WA Dec 2025 – Apr 2026

- Synthesized technical literature and investigator input for multiple NSF, SBIR, and NIH funding proposals.
- Built a Python multi-agent workflow with Google Workspace integrations to identify proposal tasks, dispatch phase-specific workers, and log results, saving an estimated four hours per week of personal coordination.